

RESEARCH ARTICLE

Assessment of ChatGPT for Engineering Statics Analysis

Benjamin Hope¹  | Jayden Bracey¹ | Sahar Choukir²  | Derek Warner^{1,2}¹Canterbury Fracture Group, Department of Civil and Natural Resources Engineering, University of Canterbury, Christchurch, New Zealand | ²Cornell Fracture Group, Civil and Environmental Engineering Department, Cornell University, Ithaca, New York, USA**Correspondence:** Sahar Choukir (sc3446@cornell.edu)**Received:** 10 February 2025 | **Revised:** 6 May 2026 | **Accepted:** 12 May 2026**Funding:** National Science Foundation, Grant/Award Number: 1922081; New Zealand Ministry of Business, Innovation, and Employment, Grant/Award Number: (UOC2454)**Keywords:** ChatGPT | engineering statics analysis | large language model | prompt engineering | undergraduate engineering education

ABSTRACT

Large language models (LLMs) such as OpenAI's ChatGPT hold potential for automating engineering analysis, yet their reliability in solving multi-step statics problems remains uncertain. This study evaluates the performance of GPT-4o and o1-preview on foundational statics tasks, from simple $F = ma$ calculations to beam and truss analyses and compares their results to first-year engineering students on a typical statics exam. To enhance accuracy, we developed a Custom GPT, embedding refined prompts directly into its instructions. This optimized model achieved performance scores ranging from 82% to 86% depending on the exam version, surpassing the 75% student average, demonstrating the impact of tailored guidance. Despite these improvements, LLMs continued to exhibit errors in nuanced or open-ended problems, such as misidentifying tension and compression in truss members. These findings highlight both the promise and current limitations of AI in structural analysis, emphasizing the need for improved reasoning, multimodal capabilities, and targeted training data for future AI-driven automation in civil and mechanical engineering.

1 | Introduction

Generative AI models leverage machine learning (ML) to analyze and synthesize large datasets, enabling the generation of contextually relevant outputs across text, images, and audio domains. Among the most transformative advancements are large language models (LLMs) such as OpenAI's ChatGPT, Anthropic's Claude, DeepSeek models and Google AI's Gemini. In academic contexts, these tools are increasingly being utilized to tackle complex problems, simplify abstract concepts, automate routine calculations, and provide immediate, personalized assistance and feedback, reshaping how students engage with learning material [1–6].

A growing body of literature has evaluated LLM performance on standardized and licensure examinations, positioning these assessments as benchmarks for measuring reasoning ability and domain knowledge. Recent studies have evaluated the

performance of LLMs on the Medical College Admission Test (MCAT) and the Scholastic Aptitude Test (SAT), and other higher-education assessments [7–13]. Following the public release of public release of GPT-4, OpenAI reported its performance across multiple academic and professional licensure exams, including the Graduate Record Examination (GRE), SAT, Law School Admission Test (LSAT), and the Uniform Bar Exam [14]. In many instances, GPT-4 scored at or above the 90th percentile compared to human test-takers.

The strongest evidence of LLM effectiveness has emerged in medical education and licensure contexts. Bommineni et al. [8] evaluated ChatGPT on a comprehensive 230-question MCAT, spanning social, physical, natural, and behavioral sciences, as well as critical reasoning. ChatGPT performed at or above the median of more than 200,000 examinees, with explanations closely aligned with official solutions. More recent evaluations

All rights reserved, including rights for text and data mining and training of artificial intelligence technologies or similar technologies.

© 2026 Wiley Periodicals LLC.

demonstrated that GPT-4o achieved a 96% accuracy rate on previously unseen medical questions and consistently outperformed medical students across six state-level examinations, with a statistically significant mean score of 95.54% compared to students' 72.15% [12]. However, it was generally found that while LLMs excelled in verbal reasoning, their performance in quantitative reasoning was notably weak [10, 15]. This limitation is particularly consequential in STEM fields, and especially in engineering, where problem-solving relies on rigorous mathematical formulation, multi-step analytical reasoning, and strict adherence to physical laws.

In fact, evidence from civil/mechanical engineering, particularly in mechanics, presents a nuanced picture and less consistent performance profile. While recent studies suggest that LLMs are beginning to demonstrate potential in applying classical mechanics theory [16–19] and supporting numerical approaches such as the finite element method [5, 20–22], their performance on foundational engineering problems remains inconsistent [23]. For example, in one evaluation of undergraduate solid mechanics coursework, ChatGPT achieved a score of only 20 out of 100, highlighting significant difficulties in executing multi-step analytical solutions despite correctly identifying relevant concepts [4, 24]. Other assessments of GPT-4o on mechanical engineering concept inventories reported best-case accuracies of approximately 71%, with performance varying widely across subtopics and question types [19]. In more calculation-intensive settings, results are notably poorer, underscoring persistent limitations.

This discrepancy raises critical questions regarding the role of LLMs in engineering education and assessment. How should we expect LLMs to perform on foundational mechanics assessments, in particular statics, which form the cornerstone of first-year civil and mechanical engineering curricula? Despite the rapid adoption of LLMs by students, systematic evaluations of their capabilities in solving engineering statics problems remain largely absent from the literature [25]. This lack of empirical evidence represents a significant gap, particularly given that statics problems require rigorous quantitative reasoning demand, strict adherence to physical law, precise free-body diagram interpretation, equilibrium formulation, and sequential mathematical multi-step reasoning. These requirements differ fundamentally from the predominantly conceptual and language-driven tasks in which LLMs currently excel.

The increasing availability of LLMs to students also raises broader ethical and pedagogical concerns [4, 26–31]. As these tools provide immediate, tailored guidance, it becomes essential to understand their impact on learning outcomes, student engagement, and assessment validity. Historical precedents, such as the integration of calculators and computers into engineering education, demonstrate that technological adoption necessitates careful reconsideration of instructional design and evaluation methods. Recent studies suggest that AI tools can enhance conceptual understanding when used appropriately. For example, Uddin et al. [6] found that civil engineering students who consulted ChatGPT exhibited improved problem-solving performance even when subsequently working without AI assistance. Similarly, instructors have noted that LLMs can explain complex engineering concepts in accessible language,

potentially accelerating students' grasp of fundamentals. Furthermore, early evaluations by Nikolic et al. [32] showed that even initial versions of ChatGPT could generate passable responses to engineering assessments, raising concerns about academic integrity and the adequacy of traditional evaluation methods. Collectively, these developments underscore the urgency of rethinking engineering assessment frameworks in response to rapidly advancing AI capabilities.

This study aims to evaluate the problem-solving accuracy and conceptual reasoning of ChatGPT models in engineering statics, foundational core course in both civil and mechanical engineering curricula. Model performance is compared with that of undergraduate students to determine the potential of LLMs as instructional and assessment tools. Specifically, the study provides a quantitative evaluation of GPT-4o and the o1-preview, on complex, multi-step statics problems representative of typical undergraduate statics assessments. Unlike prior work that emphasizes broad educational implications or high-level conceptual evaluations, this study systematically examines both final-answer correctness and the underlying step-by-step reasoning produced by the models. Multiple prompting strategies, including zero-shot, few-shot, and chain-of-thought prompting, are evaluated to assess their impact on performance. By providing a detailed, problem-level analysis of LLM performance in engineering statics, this work contributes new empirical evidence to the discussion on ethical, effective, and pedagogically sound integration of AI tools into civil and mechanical engineering education.

The remainder of this paper is organized as follows: Section 2 outlines the methodology, detailing problem selection, prompt engineering strategies, and evaluation metrics used to assess ChatGPT's performance. Section 3 examines ChatGPT's ability to solve individual engineering statics problems in detail, including basic calculations based on Newton's second law of motion and analyses of beams and trusses, and evaluates the impact of text-based and image-based prompts on its performance. Section 4 presents ChatGPT's performance on a first-year engineering statics exam, comparing it to student results and evaluating the performance of the Custom GPT designed for improved performance. It also includes a critical discussion on model biases and the integration of ChatGPT into civil and mechanical engineering education. Section 5 concludes with key insights on the role of AI in engineering mechanics education.

2 | Methodology

2.1 | ChatGPT Models

ChatGPT, developed by OpenAI, is a one of the most recognized and one of the most widely used large language model (LLM). Since its release in late 2022, it has been central to the rise of generative AI, influencing the development of others LLMs with diverse capabilities. Our research focuses specifically on evaluating the effectiveness of ChatGPT in engineering statics analysis.

Two versions of ChatGPT were assessed during this study: GPT-4o and o1-preview. GPT-4o, noted for its multimodal capabilities, can process text and images, browse the internet, and support file handling. This versatility makes it a powerful tool for a wide range

of applications, including those requiring multi-step problem-solving or interactive data analysis. Conversely, o1-preview represents OpenAI's latest advancements in LLM technology, specifically optimized for advanced reasoning tasks. It is advertised as possessing PhD-level intelligence and is reported to exhibit reduced hallucination tendencies compared to its predecessor, GPT-4o [33].

In its initial release, o1-preview was initially constrained by strict usage limits, allowing only 50 messages per week. This limited availability, combined with its higher cost, made GPT-4o the primary model used for the majority of this study due to its greater accessibility, faster response times, and fewer restrictions during the research period. However, with the full release of o1 on December 5, 2024, the usage limit was increased to 50 messages per day, and additional features, such as image uploads and analysis, were introduced.

All model queries/prompts were limited to a maximum of 4096 tokens. This is well within the 128k-token limit of GPT-4o, but near the practical max ($\approx 4k$ – $8k$ tokens) for dense evaluation tasks. Responses were checked for truncation, and if a response was incomplete, the prompt was resent up to two additional times with minor prompt adjustments to encourage a complete answer. This retry mechanism was used sparingly (fewer than 5% of cases) and only when truncation affected answer completeness.

2.2 | Temperature and Hallucination

For GPT-4o, the default temperature setting of 0.7 is used to align with typical user experiences. The temperature setting, which controls the randomness of the model's output, is critical for balancing creativity and accuracy. In contrast, for the o1-preview model, temperature and sampling parameters are internally managed and not user-adjustable. Variability in ChatGPT's responses was nevertheless observed for both models, even when identical prompts were used. For GPT-4o, this variability is attributable to the use of a non-zero temperature, which introduces stochasticity into response generation. For o1-preview, output variability arises from inherent stochastic processes within the model's internal reasoning architecture rather than explicit temperature tuning. While such stochasticity can be beneficial for open-ended tasks, it poses challenges in engineering applications where precision and repeatability are essential, particularly for problems with unique solutions.

Another significant challenge was hallucination, where the model generated incorrect or fabricated information. While hallucinations can occur even at zero temperature, though less frequently, they are more likely when the model lacks sufficient context or when the task requires reasoning beyond its training data [34–36]. This issue directly impacts the reliability of ChatGPT in engineering statics analysis, as accurate and consistent outputs are essential. While careful prompt engineering can reduce hallucinations and improve repeatability, it cannot fully eliminate these issues.

Overall, the study acknowledged these limitations and focused on optimizing prompts to mitigate some of the issues arising from temperature and hallucination, ensuring outputs were as reliable as possible for the engineering problems evaluated.

2.3 | Categorization of Errors by ChatGPT

During the evaluation of ChatGPT models on engineering statics problems, errors consistently fell into three main categories: (1) Conceptual Misinterpretation and Procedural Errors, (2) Computational Mistakes, and (3) Graphical Misrepresentation. These categories capture distinct aspects of model performance, from misunderstandings of fundamental principles to flaws in numerical execution and visual representation. Understanding these different categories of errors helps identify critical areas for future improvements in LLM's development, particularly in enhancing conceptual clarity, procedural consistency, computational reliability, and multimodal integration capabilities. In the sections below, we define each error type and provide illustrative examples using ChatGPT prompts alongside the corresponding model outputs.

2.3.1 | Conceptual Misinterpretation and Procedural Errors

Conceptual misinterpretation errors arise when ChatGPT fails to correctly apply the fundamental physical principles governing statics problems. They arise from a misinterpretation of the interplay between applied loads, support conditions, and idealized structural behavior. For instance, in truss analysis, the model may misclassify members as being in tension or compression based on incorrect assumptions about force transmission, or it may misinterpret idealizations such as pin-jointed connections, zero-force members, or joint loading constraints. A common mistake is failing to recognize that, under ideal conditions, loads in trusses should be applied only at joints and that members carry only axial forces. Errors also occur when the model overlooks structural stability or misjudges whether a system is statically determinate. These issues indicate a deficient conceptual model, leading to globally invalid solutions even if the mathematical steps themselves appear formally correct.

Procedural errors, by contrast, arise when ChatGPT displays an appropriate conceptual understanding of the statics problem but fails to execute the solution method in a consistent or accurate manner. These errors are rooted in the flawed application of known techniques and include issues such as omitting forces in free-body diagrams, applying equilibrium equations incompletely, inconsistently handling sign conventions, or prematurely assuming symmetry in reactions. For instance, the model may correctly recognize the types of supports and loading conditions in a beam problem, but neglect to include all reaction components in the moment equation or misplace a force direction when summing forces. In truss problems, procedural errors might manifest as starting the method of joints at a node with more unknowns than equations or skipping steps in force resolution. While these errors may not indicate a flawed conceptual model, they undermine the reliability of the analysis and point to a need for improved methodical consistency and structured problem-solving strategies. Notably, procedural and conceptual errors often co-occur, particularly when a flawed assumption feeds into an incorrect execution, further complicating error diagnosis. In practice, conceptual and procedural errors often blur, especially in the context of statics problems where assumptions, methods, and execution are closely intertwined.

An illustrative example of procedural error is presented in the problem shown in Figure 1. In this case, GPT-4o incorrectly

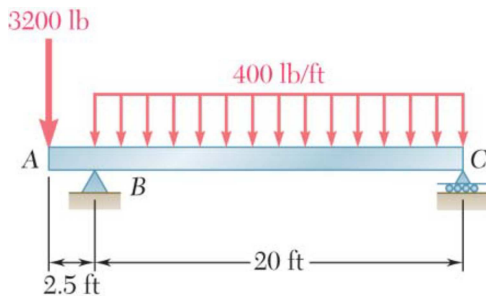


FIGURE 1 | Beam problem presented to ChatGPT [37] as a text prompt.

predicts the vertical reaction forces at supports B and C as $R_B = 6800\text{lb}$ and $R_C = 4400\text{lb}$, while the correct values are $R_B = 7600\text{lb}$ and $R_C = 3600\text{lb}$. On the procedural side, the model correctly recognizes that the uniformly distributed load (UDL) can be replaced by an equivalent point load of 8000lb acting at the midpoint of the loaded segment and correctly sets up the vertical force balance $\sum F_y = 0 = -3200 - 8000 + R_B + R_C$. However, the error emerges in the moment calculation about point B, where ChatGPT misassigns the rotational directions of the moments caused by the point load at A and the reaction force at C. Both of these forces produce clockwise moments about point B, opposing the counterclockwise moment induced by the equivalent UDL. The correct moment equilibrium equation should be: $\sum M_B = 3200 \cdot 2.5 + R_C \cdot 20 - 8000 \cdot 10 = 0$. By incorrectly assigning the signs of the moment contributions, the model arrives at erroneous values for both reaction forces. This example reveals a procedural error in the misinterpretation of moment direction conventions, layered on top of a largely correct procedural structure. It illustrates how even a fundamentally sound method can yield incorrect results when key statics concepts are misunderstood.

Prompt: I have a beam of total length 22.5 ft. A point load of 3200 lb is applied at the left end (point A, located at 0 ft). The beam is supported by a pin support at point B (located 2.5 ft from the left end), and a roller support at point C (located at the far right end, 22.5 ft from the left end). A uniformly distributed load (UDL) of 400 lb/ft is applied only between points B and C (i.e., from 2.5 ft to 22.5 ft). Calculate the vertical reaction forces at supports B and C.

GPT-4o Output: The distributed load spans from point B ($x = 2.5$ ft) to point C ($x = 22.5$ ft). Therefore, the total length of the UDL is:

$$L_{\text{UDL}} = 22.5 - 2.5 = 20 \text{ ft}$$

The total magnitude of the load is:

$$W = w \cdot L_{\text{UDL}} = 400 \cdot 20 = 8000 \text{ lb}$$

This equivalent point load acts at the midpoint of the UDL, located at:

$$x_{\text{UDL}} = 2.5 + \frac{20}{2} = 12.5 \text{ ft}$$

Let R_B be the vertical reaction at the pin support (located at $x = 2.5$ ft), and R_C the vertical reaction at the roller support (at $x = 22.5$ ft).

Sum of vertical forces:

$$R_B + R_C = 3200 + 8000 = 11200 \text{ (Eq. 1)}$$

Sum of Moments about point B

$$-R_C \cdot (22.5 - 2.5) + 3200 \cdot (2.5 - 0) + 8000 \cdot (12.5 - 2.5) = 0$$

$$-20R_C + 8000 + 80000 = 0 \Rightarrow R_C = \frac{88000}{20} = 4400 \text{ lb}$$

Substituting back into Eq. 1:

$$R_B = 11200 - 4400 = 6800 \text{ lb}$$

2.3.2 | Computational Mistakes

Computational errors are arithmetic mistakes. These mistakes were particularly evident when ChatGPT attempted intermediate step-by-step calculations, resulting in rounding errors or incorrect arithmetic operations, especially in problems involving multiple arithmetic steps.

For instance, consider solving problem shown in Figure 2. When computing the force in the frame following the prompt below, the methodology was correct, but ChatGPT arrived at a result of $F = 917.9$ lb, while the accurate value obtained using a calculator or Python is $F = 916.7$ lb. Although this discrepancy is relatively minor, similar rounding errors in other cases may lead to significantly larger deviations.

Prompt: A force F acts at point B of a two-member frame. The component of this force along member AB is 650 lb and is directed from B toward A. The component along member BC is 500 lb and is directed from B toward C. Member AB makes an angle of 60° with the horizontal (measured from point A), and member BC makes an angle of 45° with the horizontal (measured from point C).

Determine the magnitude of the resultant force F .

GPT-4o Output: We apply the Law of Cosines to find the magnitude of the resultant force F :

$$F^2 = F_{AB}^2 + F_{BC}^2 - 2 \cdot F_{AB} \cdot F_{BC} \cdot \cos(\theta)$$

Substitute the known values:

$$F^2 = 650^2 + 500^2 - 2 \cdot 650 \cdot 500 \cdot \cos(105^\circ)$$

$$F^2 = 422,500 + 250,000 - 650,000 \cdot (-0.2588)$$

$$F^2 = 672,500 + 168,220 \approx 840,720$$

$$F = \sqrt{840,720} \approx 917.9 \text{ lb}$$

2.3.3 | Graphical Misrepresentation

Errors in graphical interpretation were prevalent when ChatGPT processed image-based prompts. These errors included misidentifying nodes, omitting critical truss members, or misrepresenting geometrical dimensions from provided diagrams. Such inaccuracies significantly compromised the reliability of visual problem-solving and emphasized the limitations of current multimodal processing capabilities within ChatGPT.

(Continues)

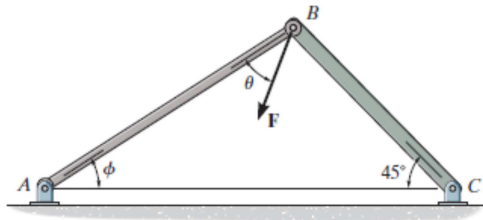


FIGURE 2 | Frame problem presented to ChatGPT [38] as a text prompt.

An illustrative example is shown below, where ChatGPT was given a high-quality image of a simple truss problem and prompted to draw the corresponding free body diagram, as seen in Figure 3. ChatGPT failed to produce a correct graphical representation. In fact, as demonstrated in Figure 4, the model misidentified the nodes (four nodes instead of three), misrepresented the angles between members and dimensions of members, and incorrectly categorized the types of supports (where it correctly recognize node C as a roller support but does not recognize node A as pin support).

Prompt: Draw the free body diagram of the truss problem shown in Figure 3.

GPT-4o Output: The generated free body diagram by ChatGPT is shown in Figure 4.

Understanding these different categories of errors helps identify critical areas for future improvements in AI model development, particularly in enhancing conceptual clarity, procedural consistency, computational reliability, and multimodal integration capabilities.

2.4 | Prompt Engineering

Prompt engineering is the systematic construction of inputs to optimize AI-generated outputs [39–43]. It can be critical to the performance of LLMs. Studies have shown that well-structured prompts can significantly improve the model's performance across various tasks [1, 44]. This study evaluated different prompting strategies to assess ChatGPT's ability to solve engineering statics problems and explored ways to enhance its performance, focusing on three key approaches: zero-shot, few-shot, and chain-of-thought (CoT) prompting.

Zero-shot, one-shot, and few-shot prompting involve providing the model with zero, one, or a few examples of how to solve a problem before asking it to tackle similar tasks. Research by Brown et al. [45] demonstrated that LLMs like ChatGPT function as effective few-shot learners, with performance improving as the number of examples in the prompt increases. In this study, these techniques were used to evaluate how ChatGPT's performance varied with different levels of guidance, revealing that including worked examples often improved the model's accuracy in solving statics problems.

Chain-of-thought (CoT) prompting further enhances performance by breaking down complex, multi-step problems into smaller, more manageable steps. This approach, shown by Wei et al. [1] to outperform standard prompting for

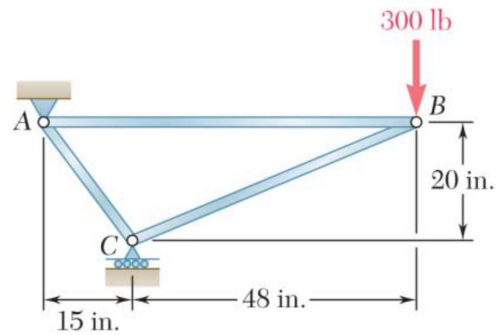


FIGURE 3 | Truss problem presented to ChatGPT [37] as a graphical prompt.

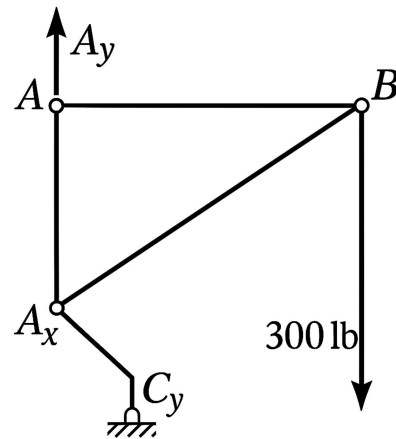


FIGURE 4 | Output of ChatGPT to Truss problem in Figure 3.

mathematical reasoning tasks, was particularly effective in improving ChatGPT's ability to tackle engineering problems requiring sequential reasoning.

To clearly illustrate the difference between these different prompting styles, let's reconsider the beam problem shown in Figure 1. The three different prompting methods are given below:

Zero-shot prompt: I have a beam of total length 22.5 ft. A point load of 3200 lb is applied at the left end (point A, located at 0 ft). The beam is supported by a pin support at point B (located 2.5 ft from the left end), and a roller support at point C (located at the far right end, 22.5 ft from the left end). A uniformly distributed load (UDL) of 400 lb/ft is applied only between points B and C (i.e., from 2.5 ft to 22.5 ft). Calculate the vertical reaction forces at supports B and C.

Few-shot prompt:

Example 1

Input: A simply supported beam is 10 ft long, with a pin at A (left) and a roller at B (right). A 1000 lb point load is applied 4 ft from A.

Output: Let R_A and R_B be the vertical reaction forces.

– Take moments about A:

$$R_B \times 10 = 1000 \times 4 \Rightarrow R_B = 400 \text{ lb}$$

– Use vertical equilibrium:

(Continues)

$$R_A + R_B = 1000 \Rightarrow R_A = 600 \text{ lb}$$

Example 2

Input: A 12 ft beam has a pin at A and a roller at B. A uniform load of 200 lb/ft acts along the entire beam.

Output: Total UDL = $200 \times 12 = 2400$ lb, acts at midpoint (6 ft).

By symmetry, $R_A = R_B = 1200$ lb

Now solve this:

Input: I have a beam of total length 22.5 ft. A point load of 3200 lb is applied at the left end (point A, located at 0 ft). The beam is supported by a pin support at point B (located 2.5 ft from the left end), and a roller support at point C (located at the far right end, 22.5 ft from the left end). A uniformly distributed load (UDL) of 400 lb/ft is applied only between points B and C (i.e., from 2.5 ft to 22.5 ft). Calculate the vertical reaction forces at supports B and C.

Chain-of-thought (CoT) prompt:

I have a beam of total length 22.5 ft. A point load of 3200 lb is applied at the far left end (point A, at 0 ft). The beam is supported by a **pin at B** (2.5 ft from the left) and a **roller at C** (22.5 ft from the left). A uniformly distributed load (UDL) of 400 lb/ft acts only between points B and C.

Let's solve this step by step:

- **Identify the external forces:**
 - 3200 lb point load at 0 ft
 - UDL of 400 lb/ft from 2.5 ft to 22.5 ft (total length = 20 ft)
 - Reaction forces: R_B at 2.5 ft (pin support), R_C at 22.5 ft (roller support)

- **Calculate the total UDL force:**

$$w = 400 \text{ lb/ft}, L = 20 \text{ ft} \Rightarrow wL = 8000 \text{ lb}$$

- **Determine the location of the UDL resultant:**

Acts at midpoint of UDL segment: $2.5 + 10 = 12.5$ ft

- **Apply moment equilibrium about point C to solve for R_B :**

$$\sum M_C = 0 \Rightarrow 3200 \times (20 + 2.5) + 8000 \times 10 - R_B \times 20 = 0$$

- **Solve for R_C ,** then use vertical force equilibrium to find R_B :

$$\sum F_y = 0 \Rightarrow R_B + R_C = 3200 + 8000 = 11200 \text{ lb}$$

- **Now complete the calculations and report R_B and R_C .**

While advanced prompting strategies such as few-shot and CoT prompting can significantly improve the performance of large language models, they also introduce a critical dependency on the user's expertise. These approaches require the user to not only understand the underlying problem but also to construct accurate, pedagogically sound prompts. In the context of engineering statics, this means selecting representative examples, ensuring correct application of physical principles, and presenting logically consistent step-by-step reasoning. Without such

foundational knowledge often lacking in junior undergraduate students, users risk introducing conceptual or procedural errors into the prompt itself, which can mislead the model and degrade its performance. In contrast, zero-shot prompting, where the model is given only the task description with no prior examples or structured reasoning, places less burden on the user and does not require any domain-specific knowledge. This eliminates the risk of cascading errors from flawed prompt construction. Therefore, the effectiveness of few-shot and CoT prompting is contingent not only on model capability but also on the user's ability to scaffold the task appropriately.

Combining few-shot and chain-of-thought (CoT) prompting techniques has been shown to improve the reliability and accuracy of large language models like ChatGPT [44]. In this study, these combined techniques were used to provide structured guidance and step-by-step reasoning for engineering statics problems. The integration of examples (few-shot) with explicit reasoning processes (CoT) aimed to improve performance by reducing errors arising from misinterpretation or inconsistent logic. Notably, ChatGPT's performance did not consistently depend on the complexity of the problem but was influenced more by the clarity and structure of the prompts provided. These combined approaches were iteratively refined to maximize accuracy and consistency across a range of problems, regardless of their complexity.

2.5 | Evaluation of Text and Image-Based Prompts

The study evaluated both text-based and image-based prompts. Text-based prompts involved detailed problem descriptions, while image-based prompts tested the models' ability to interpret diagrams and integrate visual information into their problem-solving processes. While GPT-4o supported multimodal inputs, initial findings revealed challenges in interpreting complex graphics, prompting a primary focus on text-based scenarios.

2.6 | Engineering Statics Problem Set

This study aimed to assess the reliability and pedagogical utility of large language models (LLMs), specifically GPT-4o, o1-preview, and a Custom GPT, in solving engineering statics problems from first-year civil and mechanical engineering mechanics and statics courses, including:

- **Equilibrium problems:** Determining reaction forces for simply supported beams under various loads.
- **Truss analysis:** Solving for forces in members of simple trusses subjected to point loads.
- **Beam mechanics:** Analyzing bending moments, shear forces, and deflections.
- **First-year engineering statics test:** A combination of problems mentioned above in addition to frames and 3D equilibrium problems in cables.

The evaluation encompassed a range of tasks, from basic arithmetic calculation of Newton's second law to more complex analyses involving beams, trusses, frames, and 3D equilibrium.

Final exams are typically designed to assess the full spectrum of fundamental concepts that students are expected to master by the end of a Mechanics I (Statics) course. Accordingly, the selected problems span a comprehensive set of core topics: vector force resolution and equilibrium conditions, truss analysis, beam under distributed forces, internal force components in rigid frames, and tension in cables within 3D force systems. These topics align with ABET (Accreditation Board for Engineering and Technology)-accredited course outcomes, which emphasize the construction of free-body diagrams, application of equilibrium equations, and the resolution of forces in two and three dimensions, as well as the identification of tension and compression in structural members.

The selected questions also reflect a range of cognitive demands, from basic arithmetic and procedural problem-solving to multi-step equilibrium analysis and spatial reasoning. Importantly, they are authentic assessment items, commonly used in instruction and widely featured in canonical textbooks such as Hibbeler [38, 46] and Beer and Johnston [37, 47], thereby supporting their pedagogical validity and curricular relevance. Nevertheless, this benchmark set is not without limitations. It represents a relatively small sample of exam problems from a single institutional context. Some core topics typically covered in a Statics course, such as friction, moments of inertia, virtual work, and shear and bending moment diagrams, were not assessed in this study. A more extensive evaluation, incorporating a broader dataset of final exams from multiple institutions and instructional styles, would offer a more comprehensive assessment of LLM performance in engineering education.

3 | Engineering Mechanics Exemplary Problems

In the following, we will use the format below to present the prompts and ChatGPT outputs:

Prompt: This box is a prompt to the ChatGPT.
GPT-4o Output: Response from GPT-4o. Correct value highlighted green, incorrect value highlighted red.
o1-preview Output: Response from o1-preview. Correct value highlighted green, incorrect value highlighted red.

3.1 | Basic Calculations

To evaluate ChatGPT's ability to perform the most basic mechanics calculation, a question requiring the application of Newton's second law was selected. At this stage, only the GPT-4o model was tested. The evaluation began with the following prompt:

Prompt: Provide answers to the following questions using the following equation. $F = m \cdot a$, where F is force in newtons (N), m is mass in kilograms (kg), and a is acceleration in terms of gravity (g).
--

Next specific questions were asked, requiring the calculation of force for a given mass ($m = 4\text{kg}$) without specifying acceleration

(a). This omission was intentional, designed to observe whether the model would recognize the missing information and either request it, assume a value, or proceed incorrectly.

Prompt: Calculate F for $m = 4\text{ kg}$.

GPT-4o Output: In order to calculate the force F , we need the acceleration along with the mass m . Since you provided $m = 4\text{ kg}$, please provide the value of acceleration a .

The missing information was correctly identified by ChatGPT, which then requested clarification. This was considered the appropriate and preferred response for the evaluation.

Next, ChatGPT was instructed to calculate force over 100 iterations, with varying mass and acceleration values for each trial. The correct values were calculated using Python and compared to ChatGPT's results (Figure 5). The analysis revealed a correlation coefficient (R^2) of 0.79 between the ChatGPT outputs and the correct values, with a mean percentage difference of 13.73%.

Upon investigating the discrepancies, it was discovered that ChatGPT performed the calculations using chain-of-thought (CoT) reasoning, breaking the problem down into smaller steps. For example, when $m = 4\text{ kg}$ and $a = 10\text{g}$, ChatGPT produced the following steps:

GPT-4o Output: Multiply the mass by the acceleration: $10 \times 4 = 40$, Multiply 40 by 9: $40 \times 9 = 360$, Multiply 40 by 0.81: $40 \times 0.81 = 32.4$, Add the two results together: $360 + 32.4 = 392.4$.

This process demonstrated that while ChatGPT correctly applied CoT reasoning, it introduced errors in the intermediate steps of multiplication. The unnecessary separation of 9.81 into 9 and 0.81 added complexity to the calculation. This breakdown introduced avoidable rounding and computational errors, compromising the accuracy of the final output across other trials. Previous Research has also shown that CoT prompting, while

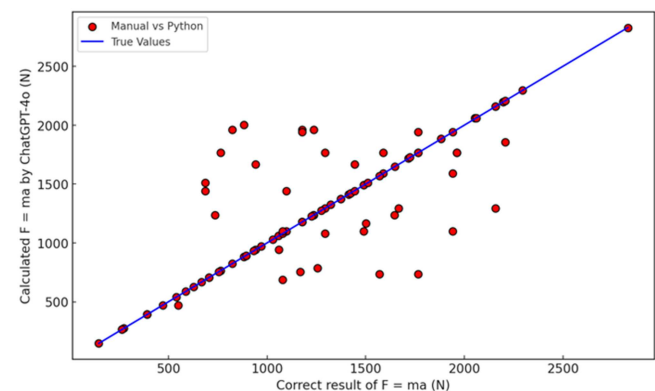


FIGURE 5 | Accuracy of ChatGPT in calculating $F = m \cdot a$ across 100 iterations. The mass (m) and acceleration (a) values were randomly generated by ChatGPT for each calculation. The y-axis represents the force (F) values calculated by ChatGPT, and the x-axis represents the correct force values calculated using Python. The blue line represents perfect accuracy.

effective for logical reasoning, is a common source of arithmetic errors [48–51]. LLMs often break calculations into unnatural sub-steps, increasing cognitive load and error rates [52].

To mitigate this issue, the original prompt was repeated, explicitly instructing ChatGPT to use Python for all calculations. Under this approach, the model generated accurate results for every trial, effectively resolving the discrepancies. This finding is consistent with recent work on program-aided language models (PAL) and chain-of-code (CoC), which demonstrate that offloading arithmetic to a deterministic interpreter dramatically improves numerical accuracy [53, 54]. By generating and executing Python code, the LLM avoids the fragile intermediate calculations that plague CoT reasoning. The instruction to use Python for all computations was maintained throughout the remainder of the research to ensure accurate calculations.

The original prompt was repeated, explicitly instructing ChatGPT to use Python for all calculations. Under this approach, the model generated accurate results for every trial, effectively resolving the discrepancies. The instruction to use Python for all computations was maintained throughout the remainder of the research to ensure accurate calculations. This finding underscores the importance of clear and specific instructions when using LLMs for numerical tasks.

3.2 | Beams and Trusses

This section examines ChatGPT's ability to solve more complex engineering statics problems, including beam reactions and truss member forces. When tasked with simpler statics problems, such as single-force or single-support scenarios, ChatGPT generally followed correct procedural steps. For instance, in truss analysis, it accurately applied equilibrium equations to determine reactions before calculating member forces via the method of joints. However, its accuracy declined as the complexity of the problems increased, an increase in complexity often driven by the number of unknowns and equations, multiple or angled loads, extended geometry (e.g., multi-span beams or various support types), and the need for deeper theoretical reasoning. In engineering statics analysis, these factors collectively define a problem's complexity: having to decompose inclined forces into orthogonal components, keeping track of multiple support reactions, determining determinacy and stability of trusses, or recognizing zero-force members each adds layers of challenge. Consequently, when problems involved multiple forces or inclined angles, or required a thorough understanding of member behavior under tension and compression, ChatGPT's reliability diminished, underscoring the model's difficulty in handling multi-step equilibrium analyses with higher conceptual demands.

Beam and truss examples are discussed in detail in Sections 3.2-1 and 3.2-2, respectively, as they both contained common issues and nuances that ChatGPT struggled to address:

- Difficulty in correctly resolving angled forces or members into their horizontal and vertical components.
- Errors in determining whether members were in tension or compression and in properly accounting for the direction of vectors when applying static equilibrium equations.

- Instances of hallucination, where the model introduced non-existent forces while using the method of joints or sections. This issue occurred more frequently when using multi-modal prompts compared to text-based prompts.

3.2.1 | Beam Example

The following zero-shot prompt was first provided to the ChatGPT models to solve the beam problem shown in Figure 6:

Prompt: I have a 6 m long beam. Point A has a roller support located at $x = 2$ m. Point B has a pinned support located at $x = 6$ m. Point C is located at $x = 4$ m. A 20 kN/m UDL is applied between $x = 0$ and $x = 2$ m. Determine the reactions at the supports, and the resultant normal force, shear force, and bending moment at Point C in the beam. Use Python for all calculations.

GPT-4o Output: Reaction at A: 10 kN, Reaction at B: 30 kN, Shear Force at C: 10 kN, Bending Moment at C: -20 kNm, Normal Force at C: 0 kN.

o1-preview Output: Reaction at A: 30 kN, Reaction at B: 10 kN, Horizontal Reaction at B: 0 kN, Shear Force at C: -10 kN, Bending Moment at C: 60 kNm, Normal Force at C: 0 kN.

Both models failed to correctly calculate the reactions due to incorrect assumptions about the problem's constraints. Due to the loading on the beam to the left of support A, the beam should rotate and bend about support A, causing a negative reaction at support B (holding the beam down). The correct reactions are 50 kN at A and -10 kN at B. Interestingly, GPT-4o correctly solved the forces at Point C, which appeared to result from a coincidence of following the correct analysis process despite the incorrect reactions. In contrast, o1-preview did not follow standard procedures for calculating shear forces and bending moments, instead breaking the beam into segments and summing the results, which led to incorrect forces.

Prompt: Break this problem down step by step. 1) Solve for the reactions at the supports. Consider how the applied forces will create bending and rotation around supports and how that might influence the reactions at the supports. 2) Solve for the forces at point C using standard analysis procedures. 3) Review your calculations and answers, check that signs (+ and -) have been correctly applied, and if your answers make logical sense (i.e. equilibrium is satisfied).

GPT-4o was further tested with a one, two, and three-shot CoT prompt, where each provided example became progressively more similar to the original question, with the third shot being nearly identical except for a different load magnitude.

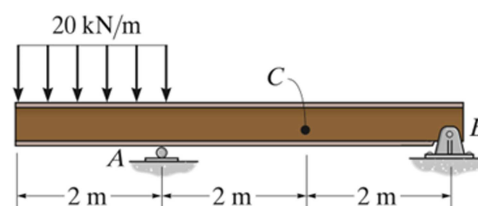


FIGURE 6 | Beam problem presented to ChatGPT [46].

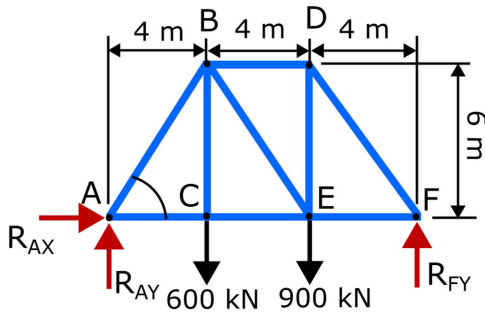


FIGURE 7 | Truss problem presented to ChatGPT [55].

GPT-4o was unable to solve the reactions correctly, and even after being provided with a nearly identical example to the original prompt, it continued to produce incorrect results for the reactions. This suggests that GPT-4o was not effectively learning or adapting from the provided inputs.

3.2.2 | Truss Example

A zero-shot prompt describing the truss problem shown in Figure 7 was first provided to the ChatGPT models:

Prompt: Q1: I have a 10 m long beam, Point A has a pinned support at $x = 0$, Point B has a roller support at $x = 5$ m, Point C is located at $x = 2$ m. A 5 kN/m UDL is applied between $x = 5$ and $x = 10$. Determine the reactions at the supports, and the resultant normal force, shear force, and bending moment at point C in the beam. **A1:** First, identifying that due to the loading of the beam to the right of support B, the beam will rotate and bend around support B, causing uplift and therefore a downwards reaction at support A. Sum of moments around point a (ACW = positive): $0 = 5B_y - 7.5(5 \times 5)$, solving for B_y :

$B_y = \frac{7.5 \times 25}{5} = 37.5$ kNm. Sum of forces in the vertical direction (up = positive): $0 = B_y - A_y - 5 \times 5$, so $A_y = B_y - 25 = 37.5 - 25 = 12.5$ kN. From the Shear Force Diagram (SFD) and Bending Moment Diagram (BMD): $V_c = -12.5$ kN, $M_c = -25$ kNm, $N_c = 0$.

Q2: A 5 m long beam with Point A as a pinned support at $x = 5$ m, Point B as a roller support at $x = 3$ m, and Point C located at $x = 4$ m. A UDL of 2 kN/m is applied between $x = 0$ and $x = 3$ m. **A2:** First, by identifying that due to the loading of the beam to the left of support B, the beam will rotate and bend about support B, causing uplift to the right of support B, and therefore the reaction at support A is negative, holding the beam down. Sum of moments around point B (ACW = positive): $0 = 1.5(2 \times 3) - 2 \cdot A_y$, so $A_y = \frac{9}{2} = 4.5$ kNm. Sum of forces in the vertical direction (up = positive): $0 = B_y - A_y - 2 \times 3$, so $B_y = A_y + 6 = 4.5 + 6 = 10.5$ kN. From the Shear Force Diagram (SFD) and Bending Moment Diagram (BMD): $V_c = 4.5$ kN, $M_c = -4.5$ kNm, $N_c = 0$.

Q3: I have a 6 m long beam, Point A has a pinned support at $x = 6$ m, Point B has a roller support at $x = 2$ m, Point C is located at $x = 4$ m. A 10 kN/m UDL is applied between $x = 0$ and $x = 2$ m. Determine the reactions at the supports,

(Continues)

and the resultant normal force, shear force, and bending moment at Point C in the beam. **A3:** First, by inspection, the loading of the beam to the left of support B will make the beam want to rotate anticlockwise and bend around support B. This will cause uplift at Point A, and therefore the reaction at support A is negative, holding the beam down. Sum of moments around Point B (ACW = positive):

$$0 = 1(2 \times 10) - 4 \cdot A_y \cdot A_y = \frac{20}{4} = 5 \text{ kNm (acting downwards).}$$

Sum of forces in the vertical direction (up = positive):

$$0 = B_y - A_y - 2 \times 10 \Rightarrow B_y = A_y + 20 = 5 + 20 = 25 \text{ kN.}$$

Drawing the Shear Force Diagram (SFD) and Bending Moment Diagram (BMD), I was able to find that $V_c = 5$ kN, $M_c = -20$ kNm, $N_c = 0$.

Q4: *original prompt with CoT*

The correct answers for the truss problem are as follows:

$$R_{AY} = 700 \text{ kN}, F_{AB} = -841.4 \text{ kN(C)}, F_{CE} = 466.8 \text{ kN(T)},$$

$$F_{DF} = -961.6 \text{ kN(C)}, R_{AX} = 0, F_{AC} = 466.8 \text{ kN(T)},$$

$$F_{BE} = 120.2 \text{ kN(T)}, F_{DE} = 800 \text{ kN(T)}, R_{FY} = 800 \text{ kN},$$

$$F_{BC} = 600 \text{ kN(T)}, F_{BD} = -533.5 \text{ kN(C)}, F_{EF} = 533.5 \text{ kN(T)}.$$

Prompt: I have a truss with the following nodes with (X, Y) coordinates: A: (0, 0), B: (4, 6), C: (4, 0), D: (8, 6), E: (8, 0), and F: (12, 0). Members connect nodes AB, AC, BC, BD, BE, CE, DE, DF, and EF. A 600 kN force is applied in the vertical (Y) direction at node C, and a 900 kN force is applied in the vertical (Y) direction at node E. The truss is supported by a pinned connection at node A, and a roller support at node F. Find the reactions at the supports and solve for the forces in each member, stating if it is in compression (C) or tension (T). Use Python for all calculations.

GPT-4o Output: $R_{Ax} = 0$ kN, $R_{Ay} = 700$ kN, $R_{Fy} = 800$ kN, Member AB: 841. 3 kN (T), Member AC: -466. 7 kN (C), Member BC: -600 kN (C), Member BD: -533. 3 kN (C), Member BE: 120. 2 kN (T), Member CE: 466. 7 kN (T), Member DE: 0 (No force), Member DF: 295. 8 kN (T), Member EF: -44. 4 kN (C)

o1-preview Output: $R_{Ax} = 0$ kN, $R_{Ay} = 700$ kN, $R_{Fy} = 800$ kN, Member AB: -841. 63 kN (C), Member AC: 467. 89 kN (T), Member BC: 600 kN (T), Member BE: -1081. 06 kN (C), Member CE: -467. 89 kN (C), Member EF: 1067. 89 kN (T), Member DF: Cannot be accurately determined with this method. BD and DE are likely zero-force members due to their configuration and the lack of loads at nodes B and D in the vertical direction.

Both models successfully determined the reactions at the supports and calculated some member forces accurately; however, significant errors were observed in the majority of member force calculations. A recurring issue was the misidentification

of whether members were in tension or compression, even when the numerical values of the forces were correct. This misclassification often propagated through the analysis, resulting in errors when summing forces at subsequent nodes. Despite these inaccuracies, both models demonstrated a fundamental understanding of the method of joints and adhered to its procedural framework.

Initially, both models attempted to write comprehensive Python code to solve the entire truss. This approach led to incorrect results, as the generated code failed to accurately account for the problem's nuances and constraints. To address this, the prompt was revised to clarify that Python should be used only for individual calculations rather than for automating the full truss analysis.

The amended prompt incorporated additional guidance, including detailed instructions, step-by-step examples, and chain-of-thought (CoT) reasoning techniques, to improve the models' performance and alignment with established engineering methodologies as shown below.

Prompt: I have a truss with the following nodes...

...Write all formulas and equations out in full and use Python to solve completed equations, do not write a code to solve the full truss. Use the following convention: Forces pointing away from the node are tension and forces pointing towards the node are compression. When solving by sum of forces in x or y direction, consider the force sign relative to the node (example: if the force is in tension and coming from the left side of the node, it should be considered negative, but if a force is coming from the right side of the node, and is also in tension it should be considered positive. The same idea applies in the vertical). To start, assume all unknown forces are in tension (pointing away from the node).

When solving a truss, break the problem into small steps: first, solve for the reactions. Next, use the method of joints for each node by identifying the forces acting on a node and starting with nodes that have the least number of unknowns. For example, consider node A: forces acting on node A include F_{AB} , F_{AC} , and R_{AY} . Summing forces in the y-direction: $R_{AY} + F_{AB} \sin(\theta) = 0F_{AB} = -\frac{R_{AY}}{\sin(\theta)}$. Since F_{AB} is negative, it points toward the node, indicating compression. Summing forces in the x-direction: $F_{AC} - F_{AB} \cos(\theta) = 0F_{AC} = F_{AB} \cos(\theta)$. Since F_{AC} is positive, it points away from the node, confirming the initial tension assumption.

With the of additional prompt details, o1-preview accurately calculated all force magnitudes but incorrectly identified that F_{BC} was in compression. Similarly, the GPT-4o model exhibited improved performance yet continued to produce errors in the force calculation, primarily due to misidentifying whether members were in tension and compression. Even after providing both models with two- and three-shot examples for node calculations, neither was able to correctly classify F_{BC} as being in tension. This issue persisted despite presenting the complete formula and variable placeholders required for summing the vertical forces at node C, highlighting a limitation in the

models' ability to interpret and apply mechanical equilibrium principles accurately.

The beam and truss findings are broadly consistent with the emerging literature on LLMs in engineering and physics education. The severity of these limitations is underscored by GenC's finding that GPT-4 achieved only 14% accuracy on the Statics portion of a standardized Fundamentals of Engineering Civil exam [56]. Such performance is far below any threshold for unsupervised engineering application. Orris and Ssegawa further demonstrated that model accuracy can drop by over 11% when problem statements are adversarially perturbed, indicating a lack of robustness to linguistic variations that would be inconsequential to a human engineer [57].

3.3 | Text-Based Prompt

The challenges outlined in this study highlight that while ChatGPT can often be guided toward correct solutions, achieving correct outcomes typically requires multiple iterations, rendering the process less efficient than manual problem-solving. The implementation of few-shot and chain-of-thought (CoT) prompting techniques improved the models' accuracy. However, both models consistently exhibited difficulty in understanding the unique nuances of individual problems and demonstrated limited ability to retain and apply learned procedures across different tasks. This necessitated repeated intervention and guidance for each new problem.

Based on the trials conducted, it is evident that these models often make mistakes. This may stem from their reliance on pattern recognition, derived from training data, rather than employing deductive reasoning to systematically solve problems. It is possible that this indicates that LLMs may perform equally well on mechanics problems that are often perceived as more difficult by humans. For example, LLMs might do equally well on mechanics problems from the fourth year of a typically university engineering curriculum, provided sufficient training data. That said, problems perceived as more difficult do often require an increased number of steps and therefore could challenge LLMs in the same way that more complex beam and truss problems do.

3.4 | Image-Based Prompts

GPT-4o's ability to interpret image-based prompts was also evaluated, particularly with Truss diagrams to assess its performance in solving basic engineering statics problems. Since mechanics problems often rely on visual aids like free-body diagrams to enhance comprehension, this evaluation examined whether providing prompts as images could improve ChatGPT's accuracy and effectiveness, especially for visual learners.

Figure 7 was presented with instructions to generate a free-body diagram, identify applied loads, and compute reactions at the supports. However, as shown in Figure 8, the model struggled with accurate image interpretation. Errors included incorrect node positioning, misrepresentation of dimensions, and the omission of key truss members (e.g., BD, BE, and DE), resulting in an incomplete and inaccurate depiction.

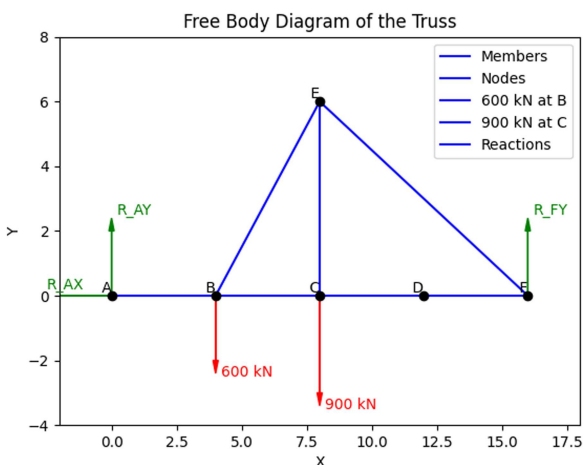


FIGURE 8 | Incorrect diagram output produced by GPT-4o from image-based prompt.

To examine these shortcomings, simplified hand-drawn sketches were subsequently utilized, omitting dimensions and text for clarity. While these modifications reduced some interpretative errors, GPT-4o continued to struggle with more complex geometries and non-standard loading scenarios, such as triangular distributed loads or inclined members. Simpler problems, such as beams with uniform distributed loads or single-point loads, were handled more effectively. These results suggest that while GPT-4o can sometimes correctly interpret simple mechanics diagrams, it often struggles with more complex diagrams.

These limitations are consistent with findings reported in prior studies. An evaluation of GPT-4o and Gemini on visual-based mechanics problems identified difficulties in accurately extracting mathematical and mechanical information from structural images [58]. More broadly, this deficiency is commonly described as “structural blindness” in multimodal large language models (MLLMs). As discussed in *Beyond Pixels*, MLLMs exhibit a critical limitation in their inability to capture topology and symbolic relationships in engineering schematics [59]. Similarly, the Math Blind study demonstrates that MLLMs perform poorly on fundamental perceptual tasks, including shape classification, object counting, relationship identification, and object grounding, often achieving near-zero accuracy in fine-grained grounding tasks [59]. The inability to correctly parse topology and symbolic logic in schematics directly explains the missing truss members and misrepresented nodes observed in Figure 8. Comparative studies further show that students consistently outperform these models on tasks requiring nuanced spatial reasoning and the interpretation of complex diagrams [60].

Collectively, these findings reinforce our results, indicating that current models remain limited in their ability to reliably interpret and reason about engineering diagrams. This limitation carries important implications for their use in mechanics education and assessment.

4 | First-Year Engineering Statics Exam

Although ChatGPT models (GPT-4o and o1-preview) make numerous errors in solving simple mechanics problems, this does not necessarily imply that their output is inferior to that of

humans. To evaluate their problem-solving capabilities in comparison to human performance, we analyzed a set of elementary mechanics problems from a first-year engineering statics exam. While ChatGPT may not truly apply engineering statics concepts, it generates solutions by identifying patterns in its training data and extrapolating from similar examples. The exam was graded consistently for both students and ChatGPT models using a common rubric, with credit awarded for the correct identification and reproduction of key principles, regardless of whether the underlying reasoning was conceptual or pattern-based.

4.1 | Evaluation Setup

Two exams were considered:

- Exam 1 served as a benchmark, enabling a comparison between student performance and ChatGPT’s GPT-4o and o1-preview models. To establish a baseline for model evaluation, the Custom ChatGPT also attempted Exam 1. Student performance was based on the responses of 705 first-year engineering students.
- Exam 2 contained similar problems and content to Exam 1 but included minor variations so that the correct answers were not identical (Figure 9). This exam was taken by 7 first-year engineering students and was used to assess the problem-solving skills of the GPT-4o, o1-preview, and Custom ChatGPT models, as well as to examine the effects of different prompt styles.

Both exams were graded using a standardized rubric to ensure consistent evaluation of student and ChatGPT responses. This uniform assessment allowed meaningful comparisons between the models and human performance, despite the disparity in participant numbers for each exam.

To investigate the influence of prompt design, seven distinct prompt styles were tested, ranging from zero-shot to chain-of-thought reasoning, with some styles incorporating structured guidance for the models. These variations were intended to expose the models’ strengths (and potential weaknesses) across a range of problem-solving scenarios.

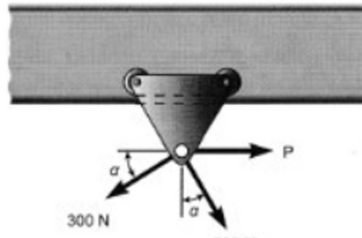
The seven distinct prompt styles are as follows:

1. Style 1: Text-based questions (TBQs) without additional instructions (zero-shot).
2. Style 2: TBQs with the instruction to “Talk me through every step and all of your thought processes,” applying chain-of-thought (CoT) reasoning.
3. Style 3: Image-based questions, tested exclusively on GPT-4o.
4. Style 4: TBQs with a specific note (Note 1) emphasizing attention to force directions, angles, and consistent sign conventions:
 - Be very careful with directions of forces and angles, keep track of these during the entire calculations, and ensure your sign convention is applied correctly.
5. Style 5: TBQs combining CoT reasoning and Note 1 instructions.

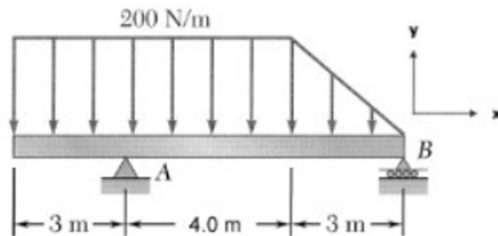
Question 1:**20 Marks**

Given: A hoist trolley is subjected to the three forces shown. Knowing that $\alpha = 30^\circ$, determine

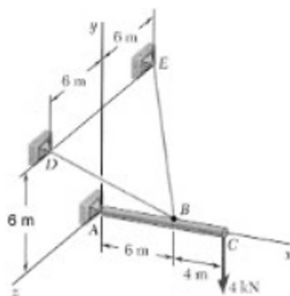
- The required magnitude of the force **P** if the resultant of the three forces is to be vertical.
- The corresponding resultant vector

**Question 3:****20 Marks**

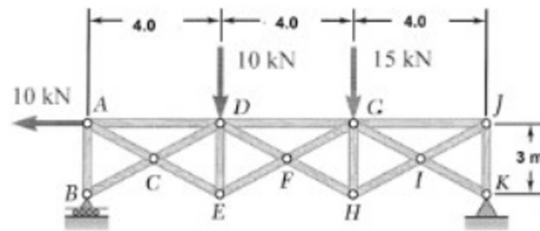
Given: Determine the reactions at the beam supports for the given load.

**Question 5:****20 Marks**

Given: A 10 m boom is acted upon by the 4 kN force shown. Determine the tension in each cable. Joint A is a ball-and-socket joint (No moments produced at A).

**Question 2:****20 Marks**

Given: The truss shown below is loaded with three forces as shown. Find the magnitude of the forces acting through the members AB, BC and IK. Indicate whether the member is in tension or compression.

**Question 4:****20 Marks**

Given: For the frame and loading shown, determine the component of all forces acting on member ABC.

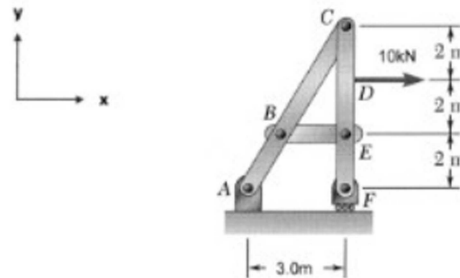


FIGURE 9 | 2024 First-year engineering statics exam referred to Exam 2. Problems adapted from [37, 47].

- Style 6: Image-based questions incorporating CoT and Note 1, tested only on GPT-4o.
- Style 7: Image-based questions with one-shot examples provided for each question, tested only on GPT-4o.

Exam 1, along with its marking rubric, was also provided to the Custom GPT model for use in solving Exam 2.

Figure 10 illustrates that, despite being advertised as having enhanced reasoning capabilities, the o1-preview model did not outperform the older model when it came to decomposing inclined forces into orthogonal components. This discrepancy is especially intriguing because one would expect a model designed with advanced reasoning features to excel in vector decomposition tasks, thereby theoretically improving performance in scenarios requiring precise breakdowns of inclined forces.

Among the tested styles, Styles 4 and 5 enabled GPT-4o to exceed the 70% threshold. Styles 2 and 5 allowed o1-preview to achieve comparable scores. Style 4, emphasizing clear instructions on sign conventions and angles, proved more effective than Style 2 (CoT) alone for reducing errors in both models. The poorest performance was observed with image-based prompts (Styles 3, 6, and 7), reinforcing the conclusion that GPT-4o's image-reading capabilities are not yet reliable for accurate engineering statics analysis.

The evaluation highlights the impact of prompt design on the performance of ChatGPT models. Style 5, which combines CoT reasoning and explicit instructions, consistently yielded the highest scores. This finding underscores the importance of detailed prompts to guide AI models in solving engineering problems accurately.

However, the limitations of the models were apparent, as reflected in their under-performance compared to students,

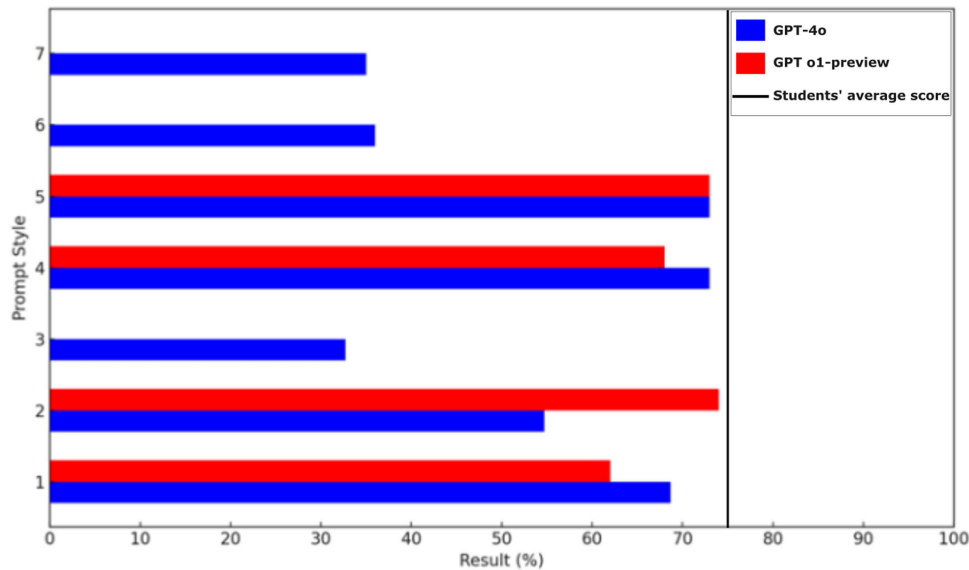


FIGURE 10 | Results comparing different prompt styles used to evaluate the GPT-4o and o1-preview models on a first-year engineering statics exam. Three styles could not be tested on GPT o1-preview due to tool limitations at the time of testing. Dark vertical line represent students' average score in Exam 1.

who scored an average of 75%. Both struggled with multi-step reasoning tasks when no additional guidance was provided, and their ability to interpret image-based prompts was inadequate. These issues suggest that while ChatGPT holds promise as a supplementary tool for solving engineering problems, it is not yet a reliable replacement for traditional methods or human expertise, particularly in multi-step scenarios.

The results of this evaluation suggest that targeted improvements in model capabilities, such as enhanced reasoning for vector decomposition and improved handling of graphical inputs, could significantly increase the utility of LLMs in engineering mechanics.

4.2 | Custom GPT

The findings from GPT-4o and o1-preview models were utilized to develop a Custom GPT model by consolidating the most effective prompts and notes identified throughout the study. This Custom GPT, accessible at <https://chatgpt.com/g/g-ZM5WXd1n2-dhw05-gpt>, incorporates detailed instructions on chain-of-thought (CoT) reasoning and specific examples designed to address challenges encountered during testing. Users can access these instructions by prompting the model with, "Provide your system prompt".

The Custom GPT was evaluated against Exam 2 using two prompt styles: Style 1 (text-based questions) and Style 3 (image-based questions), as defined in Section 4. The model achieved a score of 86% with Style 1 and 50% with Style 3. The lower performance on image-based prompts was attributed to persistent difficulties in interpreting visual information accurately, an issue noted in earlier assessments of the GPT-4o model.

The performance of first-year engineering students on these exams averaged 75% for Exam 1 and 81% for Exam 2, with 705

students taking Exam 1 and only seven taking Exam 2. Due to the small sample size for Exam 2, it was assumed that the difficulty levels of the two versions were comparable to allow for meaningful comparisons.

On Exam 2, the Custom GPT outperformed the average student, achieving a score of 86%. Similarly, on Exam 1, the model scored 82%, exceeding the class average of 75% and placing it within the top 36% of student results (Figure 11). These results highlight the potential of the Custom GPT as a supplemental educational tool, particularly in text-based applications.

The Custom GPT demonstrated notable improvements over earlier models, particularly in correctly incorporating all relevant forces in equilibrium equations, accurately determining lever arms and applying correct sign conventions and recognizing that roller supports cannot provide horizontal reaction forces.

Despite these advances, several issues persisted. In certain cases, the model incorrectly applied sign conventions, a common error among first-year students. The model occasionally assumed force directions without verification, leading to incorrect solutions. When prompted with Style 3 questions (image-based), the model struggled to process visual data effectively, often producing physically impossible or illogical outputs.

For instance, during Style 3 testing, the model incorrectly calculated a compressive force when solving for tension in a rope, failing to recognize the physical impossibility of the result. Furthermore, even when informed that the correct answer was included in its knowledge base, the model continued to produce erroneous results. This may be attributed to the prompt instructions being overly complex, potentially exceeding the model's ability to process and respond efficiently.

In the present study, ChatGPT demonstrated strong proficiency on first-year engineering statics exams, with a tailored "Custom

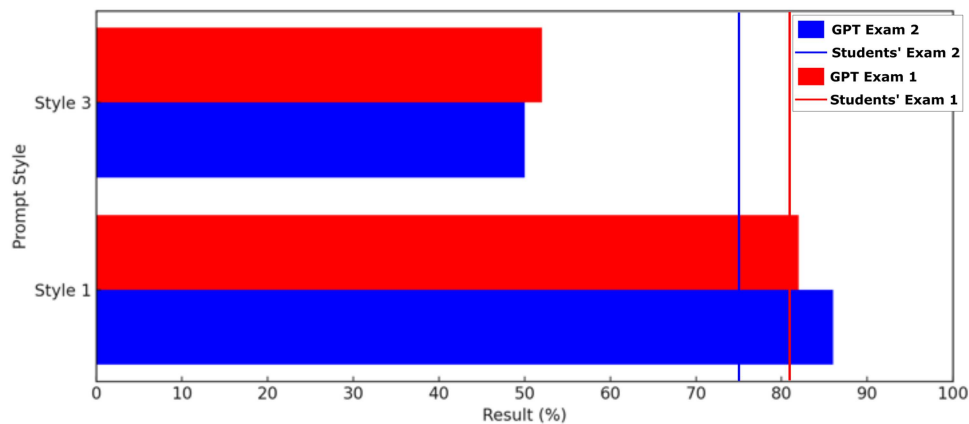


FIGURE 11 | Results comparing the Custom GPT against first-year engineering statics exams. Both the results from Exam 1 and Exam 2 are displayed. Vertical lines represent students' average score in Exam 1 (red) and in Exam 2 (blue).

GPT” model achieving scores between 82% and 86%, notably exceeding the student class average of 75%. This performance places the model within the top tier of the student cohort and highlights its ability to solve fundamental statics problems at a level comparable to high-performing human learners. However, ChatGPT's effectiveness is not consistent across all STEM domains; its performance varies significantly depending on the discipline, problem format, specific model used, and the complexity of reasoning required. For instance, in physics, ChatGPT-3.5 has been shown to struggle with conceptual understanding, scoring only 47% compared to the 70% average achieved by students, thereby underperforming relative to human learners [61]. It typically achieves only median-level scores on first-semester physics concept inventories, revealing limitations in deep reasoning tasks [61]. Similarly, in mathematics, GPT-4 performs reliably on standard undergraduate-level questions but fails to meet expectations on graduate-level problems, particularly those involving multi-step proofs or abstract reasoning [15].

In the biological and medical sciences, earlier models have also lagged behind human learners, for example, GPT-3.5 answered only 61% of questions correctly on a parasitology exam, compared to 91% accuracy by first-year medical students [62]. By contrast, GPT-4 has shown improved performance on professional biomedical assessments; in one graduate-level anatomical knowledge exam, it outscored the student cohort by a significant margin (95% vs. 72%) [12].

A 2024 study by Frenkel and Emara further explored ChatGPT's capabilities within mechanical engineering, evaluating GPT-3.5 and GPT-4 on upper-level undergraduate exams and practice problems from the Mechanical Engineering Fundamentals of Engineering (FE) exam [5]. GPT-4 achieved a score of 76%, markedly outperforming GPT-3.5's 51%. However, this performance was still insufficient to pass the full FE exam, due in part to the model's inability to interpret diagrams or visual data, critical components of many engineering problems.

These cross-disciplinary results reinforce a key insight from our own findings: despite the Custom GPT's strong overall performance on statics exams, it remains prone to critical errors when faced with multi-step reasoning problems.

Despite its improved performance on the statics Exam compared to GPT-4o and o1-preview, the Custom GPT often struggled with nuanced or open-ended problems. Persistent issues included misidentifying tension and compression in truss members and inconsistently handling inclined forces. While the model occasionally demonstrated elements of correct methodology, it rarely achieved fully accurate solutions. These limitations reflect that GPT models primarily rely on advanced pattern recognition derived from training data rather than systematically applying engineering principles.

4.3 | ChatGPT Bias and Integration in Engineering Education

While ChatGPT and similar LLMs offer promise as educational tools, their integration into engineering education must explicitly account for domain-specific biases that directly affect problem solving. ChatGPT's knowledge stems from patterns in its extensive training data, which includes text from the internet, books, and other sources. This inevitably reflects certain biases inherent in the data [63, 64]. OpenAI has not fully disclosed the composition of GPT-4's training corpus, it is widely acknowledged that English-language and Western sources dominate [65]. In statics, these biases manifest as technical and representational, including preferences in unit systems, notation, structural conventions, and canonical problem formulations.

One prominent example in statics is unit-system bias. ChatGPT frequently defaults to Imperial (US customary) units, such as pounds-force (lb), feet (ft), and kip-based loading, even when problems are stated in SI units or when no unit system is specified. This reflects the dominance of U.S.-based textbooks, engineering forums, and instructional materials in the model's training data [66]. For first-year students, such defaults can introduce confusion, dimensional inconsistency, or unnoticed unit-conversion errors, particularly in statics problems where equilibrium depends critically on consistent units.

A related bias appears in the selection of structural terminology and section naming conventions. For example, when asked to reference steel beam sections, ChatGPT often defaults to American W-shapes rather than IPE or HE sections commonly

used in European or international contexts. Similarly, truss examples and beam problems are frequently framed using US-centric design assumptions, such as pin-roller supports or standardized span lengths typical of North American pedagogy. While these conventions are not inherently incorrect, they may implicitly narrow students' exposure to global engineering practice unless instructors explicitly intervene.

Bias also emerges in problem formulation and solution strategy. ChatGPT exhibits a theoretical/textbook bias, tending to favor canonical textbook approaches and idealized conditions [67, 68]. It often immediately applies the method of joints or method of sections, without adequately checking whether the structure is statically determinate or whether alternative methods (e.g., virtual work, compatibility considerations) may be required. This reflects methodological rigidity over-indexing on common analytical methods while neglecting graphical or energy-based approaches that might be more suitable [15, 69].

Furthermore, ChatGPT demonstrates a diagram-interpretation bias, which is especially critical in statics where free-body diagrams (FBDs) are foundational. Because the model is fundamentally text-based, it struggles with spatial reasoning and may produce incomplete or incorrect FBDs, such as omitting reaction forces, misplacing vectors, or misinterpreting geometries described in prompts [70]. Our findings corroborate this limitation: both GPT-4o and o1-preview exhibited very poor performance on image-based prompts (Section 3.4). In particular, GPT-4o did not exceed 40% performance when solving statics problems based on image-based prompt styles (Section 4.1). In problems involving internal hinges, inclined supports, or spring constraints, scenarios less common in simplified textbook examples, the model may confidently generate incorrect diagrams and subsequent equilibrium equations. This reflects both the overrepresentation of idealized cases in training data and inherent weaknesses in diagram interpretation.

ChatGPT also exhibits a procedural bias toward completeness over correctness. In statics, this often manifests as confidently assigning force directions (tension vs. compression, assumed reaction directions) without verification, or proceeding with equilibrium equations even when the physical assumptions are invalid. This behavior is driven by the model's optimization for fluent, step-by-step outputs that mimic textbook solutions, even if underlying reasoning is flawed [71]. Lacking engineering judgment, the ability to assess unstated constraints or physical plausibility, the ChatGPT model can produce solutions that appear rigorous yet violate fundamental technical principles [72].

These statics-specific biases underscore the importance of explicitly teaching students to interrogate AI-generated solutions. Rather than treating ChatGPT as an authoritative solver, students should be encouraged to question unit choices, verify reaction assumptions, check determinacy, and compare AI-selected conventions against those used in their course or region. These biases can become valuable instructional tools, for example, by asking students to identify whether ChatGPT's solution implicitly assumes SI or Imperial units, or whether the selected beam section aligns with local design standards, or whether the presented method is appropriate for the problem's constraints. This critical engagement transforms AI from a

potential source of misconception into a catalyst for developing deeper professional discernment.

We offer the following recommendations, grounded in emerging research, expert analyses, and our own findings, for effectively integrating ChatGPT into mechanical and civil engineering education.

4.3.1 | *Teach the Tool, Not Just Use It*

Integrating ChatGPT into the engineering curriculum requires more than simply allowing access to the tool, it demands explicit instruction on how it works, its limitations, and its potential biases. This means not just using the tool, but teaching about the tool. Instructors can introduce a short module or discussion early in the course, or embed this content into existing first-year academic integrity workshops or engineering ethics courses, which are already common in many programs. Research supports this approach: exposing students to AI's mistakes enhances their understanding of what AI can and cannot do [16]. Through guided activities, students learn to examine outputs critically, recognize biases, and appreciate that ChatGPT does not have an inherent truth-filter [64].

This type of instruction supports broader educational goals, particularly in fostering ethical and professional responsibility. Teaching students about appropriate use, such as avoiding plagiarism and understanding academic policies, promotes transparency and accountability. Rather than banning AI tools (a strategy many find unrealistic), the goal should be to encourage responsible, informed use, reducing misuse while building essential AI literacy.

But teaching responsible use is only the first step. Students also need to learn how to use ChatGPT effectively. A key finding from our study is that prompting strategies, especially chain-of-thought (CoT) reasoning, significantly improve ChatGPT's performance and the quality of learning outcomes. This highlights the importance of introducing students to basic prompt engineering. Faculty should consider integrating short labs or workshops that demonstrate how different prompts can affect responses, when to use structured prompting methods like CoT or few-shot prompting, and how to assess the reliability of AI-generated answers.

Moreover, students should be encouraged to experiment with building their own custom GPTs, tailored to course content. This not only enhances understanding of how these tools work but also gives students greater control over how AI is applied in their learning process.

Ultimately, teaching students both how and when to use ChatGPT transforms it from a black-box answer engine into a critical thinking aid. It empowers students to use AI as a collaborative learning partner, equipping them with the skills to navigate an increasingly AI-integrated engineering landscape with confidence and discernment.

4.3.2 | *Use ChatGPT as a Learning Supplement*

ChatGPT can serve as a virtual tutor, supporting, not replacing student learning. Instructors might allow students to consult

ChatGPT after attempting a problem on their own to get feedback or hints, reinforcing concepts already taught. For instance, students might document how ChatGPT suggested an energy conservation approach and then reflect on its validity. Faculty can also require students to cite ChatGPT's input and critically assess any discrepancies. This transparency builds accountability and sharpens analytical skills.

When used thoughtfully, ChatGPT can support the development of problem-solving and soft skills [64]. Faculty should create clear guidelines, such as requiring comparison between student and AI solutions, or asking students to identify and correct errors in ChatGPT's responses. These strategies discourage over-reliance and promote critical engagement.

4.3.3 | Turn Bias into a Learning Opportunity

Given ChatGPT's Western-centric and discipline-skewed training data [65], engineering educators can turn its biases into teachable moments. For example, a class on seismic design might prompt ChatGPT and then analyze whether its answer defaults to US building codes. Such exercises prompt discussion on global engineering practices and the need to question default assumptions. In capstone or writing-intensive courses, students can analyze ChatGPT's explanations for oversimplifications or omissions and explore how context, regional, cultural, or disciplinary affects solutions.

Diversifying assignment contexts and encouraging questions like "How would this apply in a developing country?": can foster broader perspectives and critical thinking.

4.3.4 | Adapt Assessments to Mitigate Misuse

To discourage academic dishonesty, educators must rethink assessment design. AI-resistant tasks include assignments grounded in personal experience, lab data, or contextual constraints, things AI cannot fabricate convincingly. For example, analyzing a real-world structure on campus, rather than a textbook problem, ensures genuine engagement.

Oral presentations can verify understanding, and requiring students to submit their reasoning process (with any AI consultation documented) emphasizes learning over answer-getting. Occasionally, assessments might ask students to compare their answer to ChatGPT's and analyze which is better, turning AI into an object of critique rather than a shortcut.

4.3.5 | Train Educators and Establish Policies

Faculty and teaching assistants should receive training in AI tools and their limitations. Institutions can host workshops on prompt engineering, interpreting outputs, and designing AI-inclusive assignments. Departments should also publish clear policies on AI use, emphasizing citation, transparency, and student understanding. Example: "AI tools may be used for brainstorming or drafting, but all final submissions must reflect the student's own work and cite AI assistance in a footnote".

Including such policies in syllabi and honor codes frames AI tools as legitimate resources, like simulation software or calculators, with clear boundaries for use.

4.3.6 | Commit to Ongoing Evaluation

AI integration is not static. Programs should gather feedback from students and instructors to evaluate how ChatGPT is used, where it helps, and where it causes confusion. This feedback can inform iterative improvements. Departments might consider forming a standing *AI in Education* working group to monitor developments and update teaching practices accordingly.

Recent advancements include the release of specialized domain-specific GPTs such as the *Mechanical Engineering GPT* and the *Engineering GPT*. The former is trained on foundational texts such as *Shigley's Mechanical Engineering Design* and *MIL-HDBK-5J*, while the latter has been fine-tuned using civil engineering resources tailored for academic and professional contexts. These tools represent promising steps toward improving reliability and conceptual accuracy through curated, discipline-specific knowledge bases.

As AI models continue to evolve, so too must pedagogical strategies. What proves effective in 2025 may need to be re-evaluated by 2026. Staying current is essential to ensure engineering education remains relevant, rigorous, and forward-looking.

5 | Conclusions

This study assessed the capabilities of LLMs, particularly GPT-4o, o1-preview, and a Custom GPT model in solving engineering mechanics problems commonly encountered in first-year statics courses. By evaluating these models on progressively complex tasks, including simple mechanics problems including Newton's second law calculations ($F = ma$), truss and beam analyses, and a first-year statics exam, we explored their strengths and limitations.

Key findings underscore the significant influence of prompt design on model performance. Chain-of-thought (CoT) reasoning and explicit instructions, such as those included in Style 5 prompts, consistently improved accuracy, with the Custom GPT model achieving the best results. However, challenges persist in handling nuanced or open-ended problems, such as misidentifying tension and compression in truss members or inconsistently resolving inclined forces. While the models demonstrated partial correctness and occasional adherence to engineering methodologies, they seldom produced fully accurate solutions. These results are consistent with LLMs reliance on pattern recognition derived from training data, rather than systematically applying engineering principles or deductive reasoning to solve problems.

The study also highlighted the limitations of GPT-4o's image-based capabilities, which failed to improve performance in engineering statics analysis tasks. While mechanics problems are often presented with diagrams and visual aids, GPT-4o struggled to reliably interpret such prompts with the required accuracy. This limitation reinforces the need for enhanced multimodal capabilities to support visual problem-solving, a critical skill in engineering practice. Addressing these deficiencies could significantly broaden the applicability of AI tools in engineering analysis, design, and education. Targeted improvements in model capabilities, such as enhanced reasoning for vector

decomposition and improved handling of graphical inputs, could significantly boost the utility of LLMs in engineering mechanics.

Interestingly, our study highlights a parallel in LLM performance with engineering students who often struggle with fundamental quantitative questions on exams. This overlap has significant pedagogical implications. Because LLM outputs can mirror authentic student misconceptions, AI-assisted solutions in quantitative engineering courses may be considerably more difficult to detect than in text-based or coding assignments. Moreover, LLM errors tend to be systematic and reproducible; when multiple students rely on the same model, similar incorrect reasoning patterns and identical numerical errors may appear across submissions. This complicates differentiation between independent AI use and coordinated copying, increases the evidentiary burden for misconduct claims, and raises the risk of false positives when genuine student mistakes resemble AI outputs.

These findings suggest a shift in instructional strategy. Rather than focusing primarily on AI detection, engineering educators may need to proactively “stress test” assessments against LLMs during the design phase. By understanding predictable LLM failure modes, instructors can craft more robust problems that emphasize conceptual justification, structured reasoning, and deeper analytical understanding. This approach shifts academic integrity from a reactive enforcement model toward a proactive design strategy that fosters authentic learning.

Similarly, while ChatGPT models currently struggle with fully mastering the theoretical rigor required for multi-step exam questions, their ability to provide rapid, partially correct solutions suggests potential as supplemental tools for iterative analysis and design processes. Like novice engineers learning on the job, these models could become more effective with training on domain-specific datasets, enhanced reasoning capabilities, multiagent modeling, and integration into collaborative workflows.

In summary, while ChatGPT is not yet a replacement for traditional learning or expert analysis, its ability to score comparably to students on engineering exams highlights its promise in shaping the future of engineering education and incorporation in engineering practice. Future iterations of these tools could bridge the gap between theoretical problem-solving and real-world application by incorporating modular instruction design, enhanced multimodal capabilities, and specialized training in engineering principles. By leveraging these advancements, LLMs could play a vital role in equipping the next generation of engineers with the tools they need to navigate the complexities of modern design and problem-solving.

Acknowledgments

We would like to thank Hooman Pakdaman for his insightful contributions and active participation in this work. We also acknowledge the professional engineers who shared their expertise on AI applications in structural engineering, which helped refine our research objectives. Finally, we acknowledge the use of ChatGPT for its assistance in refining the language of this manuscript. This research was supported by the National Science Foundation (1922081) and the New Zealand Ministry of Business, Innovation, and Employment (UOC2454).

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

References

1. J. Wei, X. Wang, D. Schuurmans, et al., “Chain-of-Thought Prompting Elicits Reasoning in Large Language Models,” *Advances in Neural Information Processing Systems* 35 (2022): 24824–24837.
2. I. Joshi, R. Budhiraja, P. D. Tanna, et al. “With Great Power Comes Great Responsibility!”: Student and Instructor Perspectives on the Influence of LLMs on Undergraduate Engineering Education,” *arXiv e-prints*, arXiv-2309 (2023).
3. B. Borges, N. Foroutan, D. Bayazit, et al., “Could ChatGpt Get an Engineering Degree? Evaluating Higher Education Vulnerability to AI Assistants,” *Proceedings of the National Academy of Sciences* 121, no. 49 (2024): e2414955121.
4. L. White, T. Balart, S. Amani, K. J. Shryock, and K. L. Watson, “A Preliminary Exploration of The Disruption of a Generative AI Systems: Faculty/staff and Student Perceptions of ChatGPT and Its Capability of Completing Undergraduate Engineering Coursework,” *arXiv preprint arXiv:2403.01538* (2024).
5. M. E. Frenkel and H. Emara, “ChatGPT-3.5 and 4.0 and Mechanical Engineering: Examining Performance on the Fe Mechanical Engineering and Undergraduate Exams,” *Computer Applications in Engineering Education* 32 (2024): e22781.
6. S. M. J. Uddin, A. Albert, M. Tamanna, A. Ovid, and A. Alsharef, “ChatGpt as an Educational Resource for Civil Engineering Students,” *Computer Applications in Engineering Education* 32 (2024): e22747.
7. A. Gilson, C. Safranek, T. Huang, et al., “How Does ChatGPT Perform on the United States Medical Licensing Examination (USMLE)? The Implications of Large Language Models for Medical Education and Knowledge Assessment,” *JMIR Medical Education* 9 (2023): e45312.
8. V. L. Bommineni, S. Bhagwagar, D. Balcarcel, V. Bommineni, C. Davazitkos, and D. Boyer, “Performance of ChatGPT on the MCAT: The Road to Personalized and Equitable Premedical Learning,” *MedRxiv* (2023).
9. V. Pursnani, Y. Sermet, M. Kurt, and I. Demir, “Performance of ChatGPT on the US Fundamentals of Engineering Exam: Comprehensive Assessment of Proficiency and Potential Implications for Professional Environmental Engineering Practice,” *Computers and Education: Artificial Intelligence* 5 (2023): 100183.
10. L. Hickman, P. D. Dunlop, and J. L. Wolf, “The Performance of Large Language Models on Quantitative and Verbal Ability Tests: Initial Evidence and Implications for Unproctored High-Stakes Testing,” *International Journal of Selection and Assessment* 32 (2024): 499–511.
11. E. J. L. Tan, K. A. L. Ramos, M. E. K. B. Nazario, S. V. D. Lim, and S. B. Chu, *AI to the Test: Measuring ChatGPT's Objective Accuracy in the SATs in Comparison to Human Performance* (IEEE, 2024), 153–158.
12. M. Kipp, “From GPT-3.5 to GPT-4. o: A Leap in AI's Medical Exam Performance,” *Information* 15, no. 9 (2024): 543.
13. P. Newton and M. Xiromeriti, “ChatGPT Performance on Multiple Choice Question Examinations in Higher Education: A Pragmatic Scoping Review,” *Assessment & Evaluation in Higher Education* 49, no. 6 (2024): 781–798.
14. OpenAI R, *Gpt-4 Technical Report*, View in Article, Arxiv: 2303.08774 (2023).
15. S. Frieder, “Mathematical Capabilities of ChatGPT,” *arXiv preprint arXiv:2301.13867* (2023).
16. B. Huang and C. Lu, “Evaluating ChatGPT's Engineering-Reasoning Capabilities and Constraints Through Examples From Mechanical-Engineering Education,” paper presented at ASEE Conferences, Portland, Oregon, 2024, doi: 10.18260/1-2-47342.

17. G. Navas, G. Navas-Reascos, G. E. Navas-Reascos, and S. Guaño, "Innovating Engineering Education With AI: A Case Study on ChatGPT 4.0's Role in Statics-Physics," *Cogent Education* 12, no. 1 (2025): 2539545.
18. H. Buyones-Gonzalez, and M. Quezada-Espinoza, "Using ChatGPT to Improve Learning in Applied Statics for Construction Engineering Students," *2025 ASEE Annual Conference & Exposition Proceedings* (2025).
19. R. Gao, H. E. Merzdorf, X. Guo, S. Melhem, K. J. Shryock, and A. R. Srinivasa, "An Assessment of ChatGPT 4o's Performance on Mechanical Engineering Concept Inventories," *2025 ASEE Annual Conference & Exposition Proceedings* (2025).
20. G. Orlando, "Assessing ChatGPT for Coding Finite Element Methods," *Journal of Machine Learning for Modeling and Computing* 4, no. 2 (2023): 135–171.
21. L. Jose, D. Restrepo, and C. Vieira, *WIP: Supporting Student Understanding of Finite Element Analysis and Computational Science: Classroom Scaffolding and ChatGPT* (IEEE, 2024), 1–5.
22. L. Oblitas, S. Adeeb, and C. Cruz-Noguez, "Performance of ChatGPT and Gemini on the Fundamentals of Engineering (FE) Civil Exam: A Preliminary Comparison and Potential Implications for Civil Engineering Education," *SSRN* 5038555 (2024).
23. D. Tiro, *The Possibility of Applying ChatGPT (AI) for Calculations in Mechanical Engineering* (Springer, 2023), 313–320.
24. B. Ni and M. J. Buehler, "MechAgents: Large Language Model Multi-Agent Collaborations Can Solve Mechanics Problems, Generate New Data, and Integrate Knowledge," *Extreme Mechanics Letters* 67 (2024): 102131.
25. N. R. Brodnik, S. Carton, C. Muir, et al., "Perspective: Large Language Models in Applied Mechanics," *Journal of Applied Mechanics* 90, no. 10 (2023): 101008.
26. H. Else, "Abstracts Written by ChatGPT Fool Scientists," *Nature* 613 (2023): 423.
27. S. Jalil, S. Rafi, T. D. LaToza, K. Moran, and W. Lam, *ChatGPT and Software Testing Education: Promises & Perils* (IEEE, 2023), 4130–4137.
28. S. Murugesan and A. K. Cherukuri, "The Rise of Generative Artificial Intelligence and Its Impact on Education: The Promises and Perils," *Computer* 56, no. 5 (2023): 116–121.
29. M. Morsy, A. Farraj, and D. Reavis, *On the Challenges and Opportunities of Using ChatGPT in Academia* (IEEE, 2023), 1–6.
30. K. Zeeshan, T. Hämmäläinen, and P. Neittaanmäki, "ChatGPT for STEM Education: A Working Framework," *International Journal of Learning and Teaching* 10, no. 4 (2024): 544–548.
31. G. Puthumanai, T. Bretl, and M. Ornik, "The Lazy Student's Dream: ChatGPT Passing an Engineering Course on Its Own," *arXiv preprint arXiv:2503.05760* (2025).
32. S. Nikolic, S. Daniel, R. Haque, et al., "ChatGPT Versus Engineering Education Assessment: A Multidisciplinary and Multi-Institutional Benchmarking and Analysis of This Generative Artificial Intelligence Tool to Investigate Assessment Integrity," *European Journal of Engineering Education* 48, no. 4 (2023): 559–614.
33. E. Latif, Y. Zhou, S. Guo, et al., "A Systematic Assessment of OpenAI O1-Preview for Higher Order Thinking in Education," *arXiv preprint arXiv:2410.21287* (2024).
34. P. P. Ray, "ChatGPT: A Comprehensive Review on Background, Applications, Key Challenges, Bias, Ethics, Limitations and Future Scope," *Internet of Things and Cyber-Physical Systems* 3 (2023): 121–154.
35. V. Rawte, S. Chakraborty, A. Pathak, et al., "The Troubling Emergence of Hallucination in Large Language Models—An Extensive Definition, Quantification, and Prescriptive Remediations," in *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, eds. H. Bouamor, J. Pino, K. Bali (Association for Computational Linguistics, 2023), 2541–2573.
36. S. Tonmoy, S. Zaman, V. Jain, et al., "A Comprehensive Survey of Hallucination Mitigation Techniques in Large Language Models," *arXiv preprint arXiv:2401.01313* (2024).
37. F. P. Beer, E. Johnston, Jr., M. Russell, F. David, and E. R. Eisenberg, "Vector Mechanics for Engineers: Statics (SI Units)," *McGraw Hill Higher Education*, 9th Revised edition (October 2010), Capitulo 3 (2019): 75.
38. R. Hibbeler and K. B. Yap, *Statics* (Pearson, 2016).
39. H. Dang, L. Mecke, F. Lehmann, S. Goller, and D. Buschek, "How to Prompt? Opportunities and Challenges of Zero- and Few-Shot Learning for Human-AI Interaction in Creative Applications of Generative Models," *Proceedings of the 28th International Conference on Intelligent User Interfaces* (2023): 112–123.
40. S. Ekin, "Prompt Engineering for ChatGPT: A Quick Guide to Techniques, Tips, and BestPractices," *TechRxiv* (2023), <https://doi.org/10.36227/techrxiv.22683919.v2>.
41. X. Wan, R. Sun, H. Dai, S. O. Arik, and T. Pfister, "Better Zero-Shot Reasoning With Self-Adaptive Prompting," *arXiv preprint arXiv:2305.14106* (2023).
42. G. Marvin, N. Hellen, D. Jjingo, and J. Nakatumba-Nabende, *Prompt Engineering in Large Language Models* (Springer, 2023), 387–402.
43. P. Sahoo, A. K. Singh, S. Saha, V. Jain, S. Mondal, and A. Chadha, "A Systematic Survey of Prompt Engineering in Large Language Models: Techniques and Applications," *arXiv preprint arXiv:2402.07927* (2024).
44. S. Schulhoff, M. Ilie, N. Balepur, et al., "The Prompt Report: A Systematic Survey of Prompting Techniques," *arXiv preprint arXiv:2406.06608* (2024).
45. T. B. Brown, B. Mann, N. Ryder, et al., "Language Models Are Few-Shot Learners," *Advances in Neural Information Processing Systems* 33 (2020): 1877–1901.
46. R. C. Hibbeler, *Statics and Mechanics of Materials in SI Units* (Pearson Higher Ed, 2018).
47. F. Beer, E. Johnston, and J. DeWolf, "Mechanics of Materials, 5th SI Edition," *Stress* 1, no. 10 (1999): 1–12.
48. Q. Lyu, S. Havaladar, A. Stein, et al., "Faithful Chain-of-Thought Reasoning (2023), 305–329.
49. S. Amjith, M. Dusad, N. Muramalla, and S. Shah, "Can Large Reasoning Models Improve Accuracy on Mathematical Tasks Using Flawed Thinking?," *arXiv preprint arXiv:2512.17079* (2025).
50. J. Sun, Y. Pan, and X. Yan, "Improving Intermediate Reasoning in Zero-Shot Chain-of-Thought for Large Language Models With Filter Supervisor-Self Correction," *Neurocomputing* 620 (2025): 129219.
51. F. Wang, A. Alazali, and Y. Zhong, "How Does Unfaithful Reasoning Emerge from Autoregressive Training? A Study of Synthetic Experiments," *arXiv preprint arXiv:2602.01017* (2026).
52. N. Dziri, X. Lu, M. Sclar, et al., "Faith and Fate: Limits of Transformers on Compositionality," *Advances in Neural Information Processing Systems* 36 (2023): 70293–70332.
53. L. Gao, A. Madaan, S. Zhou, et al., "PAL: Program-Aided Language Models," *Proceedings of the 40th International Conference on Machine Learning, Proceedings of Machine Learning Research* 202 (2023), 10764–10799.
54. C. Li, J. Liang, A. Zeng, et al., "Chain of Code: Reasoning With a Language Model-Augmented Code Emulator," *Proceedings of the 41st International Conference on Machine Learning, Proceedings of Machine Learning Research* 235 (2024): 28259–28277.
55. C. M. Fleischmann, ENGR102 Statics Lecture 10, Lecture Delivered at University of Canterbury (2019) Lecture.
56. O. Genç, "Is Conversational AI Ready for Engineering Licensure? A ChatGPT Performance Benchmark and Prompting Strategy Evaluation," *Journal of Asian Architecture and Building Engineering* 24 (2025): 1–11.

57. C. Orris and P. Ssegawa, "Assessment of the Potential of Emerging Large Language Models in Structural Analysis: A Case Study on Beam Analysis" (Master's thesis, Makerere University, 2025).
58. Y. Zhang and R. K. L. Su, "Evaluating Performance of Large Language Models on Fundamental Structural Knowledge," *Proceedings of the Institution of Civil Engineers-Structures and Buildings* 178, no. 11 (2025): 1009–1017.
59. C. Ma, Z. Tian, Z. Zhou, et al., *Beyond Pixels: A Vector-to-Graph Framework for Reliable Schematic Auditing* (IEEE, 2026), 621–625.
60. G. Polverini and B. Gregorcic, "Evaluating Vision-Capable Chatbots in Interpreting Kinematics Graphs: A Comparative Study of Free and Subscription-Based Models," *Frontiers in Education* 9 (2024): 1452414.
61. C. K. Lo, "What Is the Impact of ChatGPT on Education? A Rapid Review of the Literature," *Education Sciences* 13, no. 4 (2023): 410.
62. S. Huh and S. Huh, "Are ChatGPT's Knowledge and Interpretation Ability Comparable to Those of Medical Students in Korea for Taking a Parasitology Examination?: A Descriptive Study," *Journal of Educational Evaluation for Health Professions* 20, no. 1 (2023): 1.
63. L. Lucy and D. Bamman, "Gender and Representation Bias in GPT-3 Generated Stories," *Proceedings of the Third Workshop on Narrative Understanding* (2021), 48–55.
64. R. Assaf, A. El-Qanni, I. H. Alsurakji, and M. Assad, "The ChatGPT Dilemma: Opportunities vs. Risks in Engineering Education," *An-Najah University Journal for Research-B (Humanities)* 38, no. 10 (2024): 1985–2002.
65. M. Li, A. Enkhtur, B. A. Yamamoto, F. Cheng, and L. Chen, "Potential Societal Biases of ChatGPT in Higher Education: A Scoping Review," *arXiv preprint arXiv:2311.14381* (2023).
66. S. L. Blodgett, S. Barocas, H. Daumé III, and H. Wallach, "Language (Technology) Is Power: A Critical Survey of "Bias" in NLP," *arXiv preprint arXiv:2005.14050* (2020).
67. S. A. Athaluri, S. V. Manthena, V. S. R. K. M. Kesapragada, V. Yarlagadda, T. Dave, and R. T. S. Duddumpudi, "Exploring the Boundaries of Reality: Investigating the Phenomenon of Artificial Intelligence Hallucination in Scientific Writing Through ChatGPT References," *Cureus* 15, no. 4 (2023): e37432.
68. E. Kasneci, K. Sessler, S. Küchemann, et al., "ChatGPT for Good? on Opportunities and Challenges of Large Language Models for Education," *Learning and Individual Differences* 103 (2023): 102274.
69. E. Davis and S. Aaronson, "Testing GPT-4 With Wolfram Alpha and Code Interpreter Plug-Ins on Math and Science Problems," *arXiv preprint arXiv:2308.05713* (2023).
70. S. Bubeck, V. Chandrasekaran, R. Eldan, et al., "Sparks of Artificial General Intelligence: Early Experiments With GPT-4," *arXiv preprint arXiv:2303.12712* (2023).
71. R. Bommasani, "On the Opportunities and Risks of Foundation Models," *arXiv preprint arXiv:2108.07258* (2021).
72. L. Weidinger, M. Rauh, N. Marchal, et al., "Sociotechnical Safety Evaluation of Generative AI Systems," *arXiv preprint arXiv:2310.11986* (2023).